

Renewable Energy Sources- overview of Nuclear, Geothermal, Tidal, Solar, Wind Energy Sources

NUCLEAR ENERGY RESOURCES

What is Nuclear Energy?

Nuclear energy is the energy that holds together the nucleus of atoms. It can be released from atoms in two ways: nuclear fusion and nuclear fission. In nuclear fusion, energy is released when atoms are combined or fused together to form a larger atom. In nuclear fission, atoms are split apart to form smaller atoms, releasing energy.

Nuclear power plants use nuclear fission to produce electricity by using the nuclear energy released in a reactor to heat water into steam which in turn will turn the turbine and generate electricity. It is carried out with extreme care and caution as very large amounts of energy is produced and the fuel is radioactive.

Nuclear energy constitutes 10% of the net world energy production (NPR 2019)

History

- On 2 December 1942, the first human-made self-sustaining nuclear chain reaction was initiated in Chicago Pile-1, during an experiment led by Enrico Fermi in Chicago, USA.
- On 1951 December 20 in Arco, Idaho, Experimental Breeder Reactor I produced the first electric power from nuclear energy, lighting four light bulbs.
- On June 26, 1954, the world's first nuclear power station to generate electricity for a power grid, the Obninsk Nuclear Power Plant, started operations in Obninsk of the Soviet Union.
- On 1955 August 8-20 in Geneva, Switzerland, hosted the first United Nations International Conference on the Peaceful Uses of Atomic Energy.
- On 1957 December 2, the world's first large-scale nuclear powerplant begins operation in Shipping port, Pennsylvania. The plant reached full power three weeks later and supplied electricity to the Pittsburgh area.
- On 26 April 1986, the Chernobyl disaster occurred at the No. 4 nuclear reactor in the Chernobyl Nuclear Power Plant, near the city of Pripyat in the north of the Ukrainian SSR
- On 11 March 2011, the Fukushima Daiichi nuclear disaster occurred at the Fukushima Daiichi Nuclear Power Plant in Ōkuma, Japan
- There are now 442 nuclear power plant units in 31 countries producing electricity of about 384 GW.
- There are now 66 plants with a capacity of 65 GW in 16 countries.

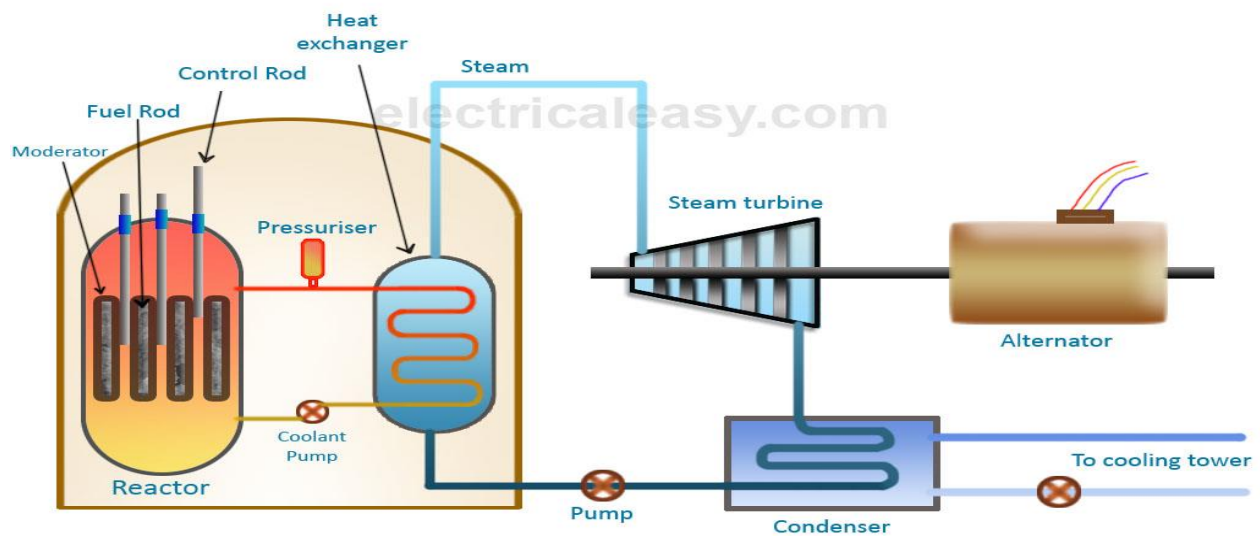
Benefits

- **Energy Production cost is competitive**
Although initial costs are high, the energy production cost is relatively less and competitive as compared to other sources
- **Nuclear energy is reliable**
Provides a stable base load and doesn't vary constantly. Also, reactor downtime for refuelling is very small.
- **Environment Friendly**
Low pollution rate than other methods of energy production and has the least carbon footprint
- **High energy Density (High efficiency)**
Amount of energy being released to the amount of fuel being used is very high
- **Sustainable and safe when properly maintained and run**
- **Nuclear power station is compact**
Geographical area of construction and functioning of the plant required is very less.

Technology: Design and Components

Basic Components and Design of Nuclear Power Plant

In a nuclear power plant, heat energy is generated by a nuclear reaction called as nuclear fission. Nuclear fission of heavy elements such as Uranium or Thorium is carried out in a special apparatus called as a nuclear reactor. Other parts of a nuclear power plant are very similar to conventional thermal power plants.



Components

- **Nuclear Reactor**

A nuclear reactor is a special apparatus used to perform nuclear fission. Since the nuclear fission is radioactive, the reactor is covered by a protective shield.

- **Heat exchanger**

In the heat exchanger, the primary coolant transfers heat to the secondary coolant (water). Thus, water from the secondary loop is converted into steam. The primary system and secondary system are closed loop, and they are never allowed to mix up with each other. Thus, heat exchanger helps in keeping secondary system free from radioactive stuff. Heat exchanger is absent in boiling water reactors.

- **Steam Turbine**

Generated steam is passed through a steam turbine, which runs due to pressure of the steam. As the steam is passed through the turbine blades, the pressure of steam gradually decreases and it expands in volume. The steam turbine is coupled to an alternator through a rotating shaft.

- **Alternator**

The steam turbine rotates the shaft of an alternator thus generating electrical energy. Electrical output of the alternator is the delivered to a step-up transformer to transfer it over distances.

- **Condenser**

The steam coming out of the turbine, after it has done its work, is then converted back into water in a condenser and pumped back to the heat exchanger.

Components of a nuclear reactor

There are several components common to most types of reactors:

- **Fuel**

Uranium is the basic fuel. Usually pellets of uranium oxide (UO_2) are arranged in tubes to form fuel rods. The rods are arranged into fuel assemblies in the reactor core.

- **Moderator**

Material in the core which slows down the neutrons released from fission so that they cause more fission. It is usually water, but may be heavy water or graphite.

- **Control rods**
These are made with neutron-absorbing material such as cadmium, hafnium or boron, and are inserted or withdrawn from the core to control the rate of reaction, or to halt it.
- **Coolant**
A fluid circulating through the core so as to transfer the heat from it. In light water reactors the water moderator functions also as primary coolant. Except in BWRs, there is secondary coolant circuit where the water becomes steam.
- **Pressure vessel or pressure tubes**
Usually a robust steel vessel containing the reactor core and moderator/coolant, but it may be a series of tubes holding the fuel and conveying the coolant through the surrounding moderator.
- **Steam generator**
Part of the cooling system of pressurised water reactors (PWR & PHWR) where the high-pressure primary coolant bringing heat from the reactor is used to make steam for the turbine, in a secondary circuit.
- **Containment**
The structure around the reactor and associated steam generators which is designed to protect it from outside intrusion and to protect those outside from the effects of radiation in case of any serious malfunction inside. It is typically a metre-thick concrete and steel structure

Economics of Nuclear Power

Nuclear power plants are expensive to build but relatively cheap to run. In many places, nuclear energy is competitive with fossil fuels as a means of electricity generation.

The economics of nuclear power involves consideration of several aspects:

- **Capital costs**, which include the cost of site preparation, construction, manufacture, commissioning and financing a nuclear power plant. Capital costs may be calculated with the financing costs included or excluded. If financing costs are included then the capital costs change materially in relation to construction time of the plant and with the interest rate and/or mode of financing employed.
- **Plant operating costs**, which include the costs of fuel, operation and maintenance (O&M), and a provision for funding the costs of decommissioning the plant and treating and disposing of used fuel and wastes. Operating costs may be divided into 'fixed costs' that are incurred whether or not the plant is generating electricity and 'variable costs', which vary in relation to the output. To calculate the operating cost of a plant over its whole lifetime (including the costs of decommissioning and used fuel and waste management), we must estimate the 'levelized' cost at present value. The levelized cost of energy (LCOE) represents the price that the electricity must fetch if the project is to break even.
- **External costs**, which are not included in the building and operation of any power plant, and are not paid by the electricity consumer, but by the community generally. The external costs are defined as those actually incurred in relation to health and the environment, and which are quantifiable but not built into the cost of the electricity.
- **Other costs** such as system costs and nuclear-specific taxes. The costs incurred in providing backup and transmission/distribution facilities are known as system costs. System costs are external to the building and operation of any power plant, but must be paid by the electricity consumer, usually as part of the transmission and distribution cost.
 - In USA, the average nuclear energy generation costs have gone down from 42/MWh in 2012 to 33/MWh in 2018.
 - In India, the cost of generation ranges between 2 Rs/KWh to 3.5 Rs/KWh.

Global and National Scenario

Global Scenario

Nuclear energy now provides about 10% of the world's electricity from about 440 power reactors. It's the world's second largest source of low-carbon power (29% of the total in 2017). In 2018, nuclear plants supplied 2563 TWh of electricity, up from 2503 TWh in 2017. This is the sixth consecutive year that global nuclear generation has risen, with output 217 TWh higher than in 2012.

Twelve countries in 2018 produced at least one-quarter of their electricity from nuclear.

United States is the world's largest producer of nuclear energy, producing 808028.33 GWh of electricity. It accounts for over 30% of worldwide nuclear energy generation. As of October 2017, the country had 99 nuclear reactors in 30 states. It produced nearly 20% of the country's total electrical output.

China has the fastest growing nuclear power program with 11 new reactors under construction. There are 47 operable nuclear reactors, with a combined net capacity of 45.7 GWe. In 2018, nuclear generated 4% of the country's electricity.

France has 58 operable nuclear reactors, with a combined net capacity of 63.1 GWe. In 2018, nuclear generated 72% of the country's electricity.

Russia has 38 operable nuclear reactors, with a combined net capacity of 29.2 GWe. In 2018, nuclear generated 18% of the country's electricity.

Tokyo Electric Power Co.'s (TEPCO) Kashiwazaki-Kariwa plant in Japan is currently the world's largest nuclear power plant, with a net capacity of 7,965MW. It has seven boiling water reactors (BWR) with a gross installed capacity of 8,212MW.

National Scenario

Nuclear power is the fifth-largest source of electricity in India after coal, gas, hydroelectricity and wind power. As of March 2018, India has 22 nuclear reactors in operation in 7 nuclear power plants, with a total installed capacity of 6,780 MW. Nuclear power produced a total of 35 TWh and supplied 3.22% of Indian electricity in 2017. 7 more reactors are under construction with a combined generation capacity of 4,300 MW. Nuclear power in India has suffered from generally low capacity factors. India's domestic uranium reserves are small and the country is dependent on uranium imports to fuel its nuclear power industry. Kazakhstan has been the largest supplier of uranium to India providing 5,000 tonnes during 2015-19.

Tarapur Atomic Power Station (T.A.P.S.) located in Tarapur, Palghar is the first commercial nuclear power station built in India. It has a capacity of 1400MW with two boiling water (BWR) and pressurised heavy water reactor (PHWR) each.

Research and Future Possibilities

- **Introduction of IV Generation reactors**

Generation IV reactors (Gen IV) are a set of nuclear reactor designs currently being researched for commercial applications by the Generation IV International Forum. They are motivated by a variety of goals including improved safety, sustainability, efficiency, and cost.

- **Fast neutron reactors (FNRs) development**

Fast neutron reactors (FNRs) can typically multiply by over a factor 50 the energy production from a given amount of uranium fuel compared to current reactors.

- **Use of Closed Fuel cycle in reactors**

Maximises the resource base and minimises high-level wastes to be sent to a repository

- **Application of Nuclear Fusion Instead of Fission**

Nuclear fusion is a reaction in which two or more atomic nuclei are combined to form one or more different atomic nuclei and subatomic particles. The difference in mass between the reactants and products is manifested as either the release or absorption of energy.

- **ITER (International Thermonuclear Experimental Reactor)**, the first fusion reactor is to be built in France by the seven collaborating partners Europe, USA, Russia, Japan, Korea, China, India

- **Waste disposal using Deep Geological Repositories**

Substantial improvements must be made with respect to the following issues:

- Primary system design simplification
- Improvement in materials
- Innovative heat exchangers and power conversion systems
- Advanced instrumentation, in-service inspection systems
- Enhanced safety

and those in fuel cycle issues pertain to:

- Improvement in partitioning and transmutation
- Innovative fuels (including minor actinide-bearing) and core performance
- Advanced separation both via aqueous processes supplementing the PUREX process as well as pyro-processing, which is mandatory for the reprocessing of the high MA-containing fuels,
- Develop a final depository.

Environmental Impact

- **Carbon emissions through related activities**
Nuclear power plants may not emit carbon dioxide during operation, but high amounts of carbon dioxide are emitted in activities related to building and running the plants. The process of mining uranium releases high amounts of carbon dioxide into the environment. Finally, the building of the power plant and transport of radioactive waste also causes carbon dioxide emissions.
- **Low Level Radiation**
Nuclear power plants constantly emit low levels of radiation into the environment. Long-term exposure to low level radiation has been shown to damage DNA. The degree of damage low levels of radiation cause to wildlife, plants and the ozone layer is not fully understood.
- **Radioactive Waste**
Radioactive waste is a huge concern. Waste from nuclear power plants can remain active for hundreds of thousands of years. Currently, much of the radioactive waste from nuclear power plants has been stored at the power plant. Due to space constraints, eventually the radioactive waste will need to be relocated.
- **Cooling Water System**
Cooling systems are used to keep nuclear power plants from overheating. There are two main environmental problems associated with nuclear power plant cooling systems. First, the cooling system pulls water from an ocean or river source. Fish are inadvertently captured in the cooling system intake and killed. Second, after the water is used to cool the power plant, it is returned to the ocean or river. The water that is returned is approximately 25 degrees warmer than the water was originally. The warmer water kills some species of fish and plant life.
- **Nuclear Power Plant Accidents and Terrorism**
According to the Union of Concerned Scientists, regulated safety procedures are not being followed to ensure that nuclear power plants are safe. Even if all safety precautions are followed, it is no guarantee that a nuclear power plant accident will not occur. Terrorism threats are another concern that needs to be addressed. A satisfactory plan to protect nuclear power plants from terrorism is not in place.

Nuclear Waste Disposal Techniques

- **Geological Disposal**
The process of geological disposal centres on burrowing nuclear waste into the ground to the point where it is out of human reach. There are a number of issues that can arise as a result of placing waste in the ground. The waste needs to be properly protected to stop any material from leaking out. Seepage from the waste could contaminate the water table if the burial location is above or below the water level. Furthermore, the waste needs to be properly fastened to the

burial site and also structurally supported in the event of a major seismic event, which could result in immediate contamination.

- **Reprocessing**

Reprocessing has also emerged as a viable long-term method for dealing with waste. This process involves taking waste and separating the useful components from those that aren't as useful. Specifically, it involves taking the fissionable material out from the irradiated nuclear fuel. Concerns regarding re-processing have largely focused around nuclear proliferation and how much easier re-processing would allow fissionable material to spread.

- **Transmutation**

Transmutation also poses a solution for long term disposal. It specifically involves converting a chemical element into another less harmful one. Common conversions include going from Chlorine to Argon or from Potassium to Argon. The driving force behind transmutation is chemical reactions that are caused from an outside stimulus, such as a proton hitting the reaction materials.

- **Space Disposal**

Space disposal has emerged as an option, but not as a very viable one. Specifically, building space disposal centres to store nuclear waste from launched space shuttles. One downhill is that the amount of nuclear waste that could be shipped on a single shuttle would be extremely small compared to the total amount of waste that would need to be dealt with. Furthermore, the possibility of the shuttle exploding enroute to space could cause the nuclear waste to spread out far beyond any reasonable measure of control. The upside would centre around the fact that launching the material into space would subvert any of the other issues associated with the other disposal methods as the decay of the material would occur outside of our atmosphere.

Geothermal Energy

What is Geothermal Energy?

The word geothermal energy comes from the Greek word Geo (earth) and therme (heat). Geothermal energy is the heat from the Earth. Its source lies 6,500 km beneath the Earth's surface, Core containing hot magma. The earth crust broken into several pieces called plates. Magma close comes to the earth surface near the edges of these plates. In these places volcano occurs. The lava that erupts from volcanos is partly magma.

Deep underground the rocks and water absorbed the heat from the magma the temperature of rocks increasing around the depth. Earth's internal heat is thermal energy generated from radioactive decay and continual heat loss from Earth's formation. Temperatures at the core–mantle boundary may reach over 4000 °C (7,200 °F). The high temperature and pressure in Earth's interior cause some rock to melt and solid mantle to behave plastically, resulting in portions of the mantle convecting upward since it is lighter than the surrounding rock. Rock and water is heated in the crust, sometimes up to 370 °C (700 °F).

History of Geothermal Energy

The use of geothermal thermal energy for heating surface is not new. Ancient people used it for heating and bathing through hot springs.

Using geothermal energy to produce electricity is a new industry. A group of Italians first used it in 1904. The Italians used the natural steam erupting from the Earth to power a turbine generator. Many ancient peoples, including the Romans, Chinese, and Native Americans, used hot mineral springs for bathing, cooking, and heating. Water from hot springs is now used world-wide in spas, for heating buildings, and for agricultural and industrial uses. Many people believe hot mineral springs have natural healing powers. Using geothermal energy to produce electricity is a relatively new industry. It was initiated by a group of Italians who built an electric generator at Lardarello in 1904. Their generator was powered by the natural steam erupting from the earth. The first attempt to develop geothermal power in the United States came in 1922 at The Geysers steam field in northern California. The project failed because the pipes and turbines of the day could not stand up to the abrasion and corrosion of the particles and impurities that were in the steam. Later, a small but successful hydrothermal plant opened at the Geysers in 1960. Today 28 plants are operating there.

Benefits of Geothermal Energy

- Clean and Green Energy Source

This is the key features of geothermal energy. Geothermal Energy sources create very little or no pollution comparing to the other mode energy production.

- Unlimited and renewable supply

Unlike Other sources of renewable energy sources like solar energy, wind energy, Tidal energy Geothermal Energy is not affected by the seasonal changes. Once The power plant is erected it get continuous and steady supply of heat from the earth.

- Available in everywhere of the world

Unlike other forms of energy which are distributed in some areas, Geothermal Energy is distributed somewhat equally around the world. New Techniques like Enhanced Geothermal System (EGS) Promises to make geothermal energy available to anywhere in the world irrespective of active volcanic activities.

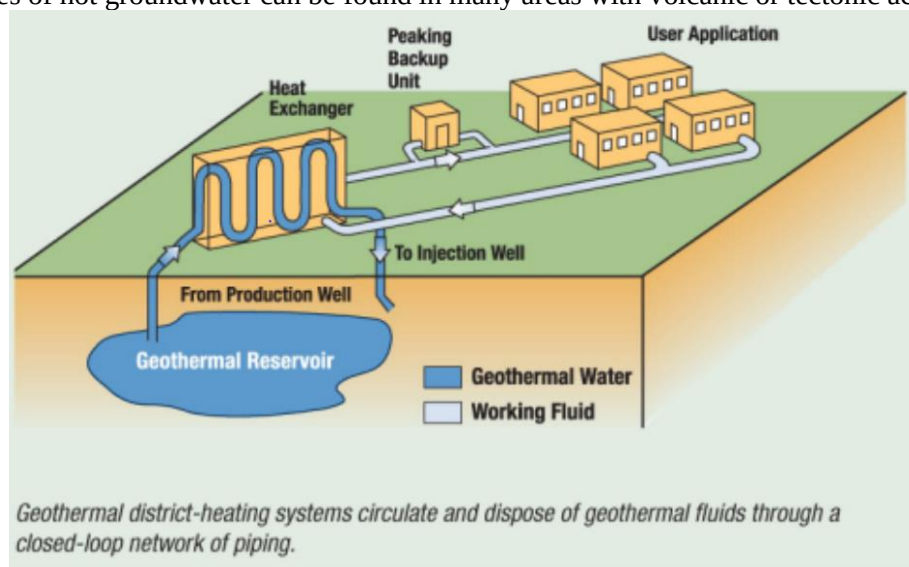
- Energy production is not affected by seasons
- Low carbon footprint when compared with fossil fuels
- Small area of land is needed
- No power storage is required
- One of the cheapest forms of energy that exists

Economics of Geothermal Energy

Unlike traditional power plants that run on fuel that must be purchased over the life of the plant, Geothermal Power plants use a renewable resource that is not susceptible to price fluctuations. Most of the cost related to geothermal power plants are related to resource exploration and plant construction. The risk in drilling of a wells is very high (one out of five well fails while drilling). This raises the cost related to resource exploration considerably. Over the last 20 years capital cost of geothermal power systems decreased by a significant 50 %. Even though India has a potential of 10600 MW geothermal capacity, availability of coal at cheap cost drags development in this field in India. Increasing environmental problems is a concern for India as a developing country. This will eventually make India depend on renewable energy sources one of which could be geothermal energy.

Technologies Used

Three main types of technologies take advantage of Earth as a heat source are. Ground source heat pumps, Direct use geothermal, Deep and enhanced geothermal systems. A ground source heat pump takes advantage of the naturally occurring difference between the above-ground air temperature and the subsurface soil temperature to move heat in support of end uses such as space heating, space cooling (air conditioning), and even water heating. A ground source or geo-exchange system consists of a heat pump connected to a series of buried pipes. Direct use of geothermal systems uses groundwater that is heated by natural geological processes below the Earth's surface. This water can be as hot as 200°F or more. Bodies of hot groundwater can be found in many areas with volcanic or tectonic activity.

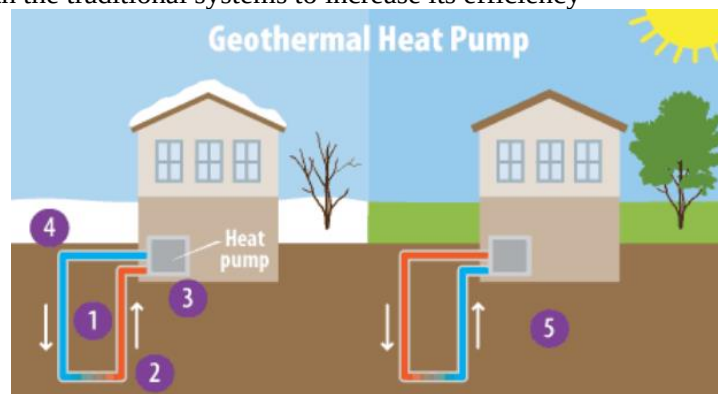


Deep geothermal systems use steam from far below the Earth's surface for applications that require temperatures of several hundred degrees Fahrenheit. These systems typically inject water into the ground through one well and bring water or steam to the surface through another. Other variations can capture steam directly from the underground ("dry steam"). Unlike ground source heat pumps or direct use geothermal systems, deep geothermal projects can involve drilling a mile or more below the Earth's surface. At these depths, high pressure keeps the water in a liquid state even at temperatures of several hundred degrees Fahrenheit.

Geothermal Heat Pumps

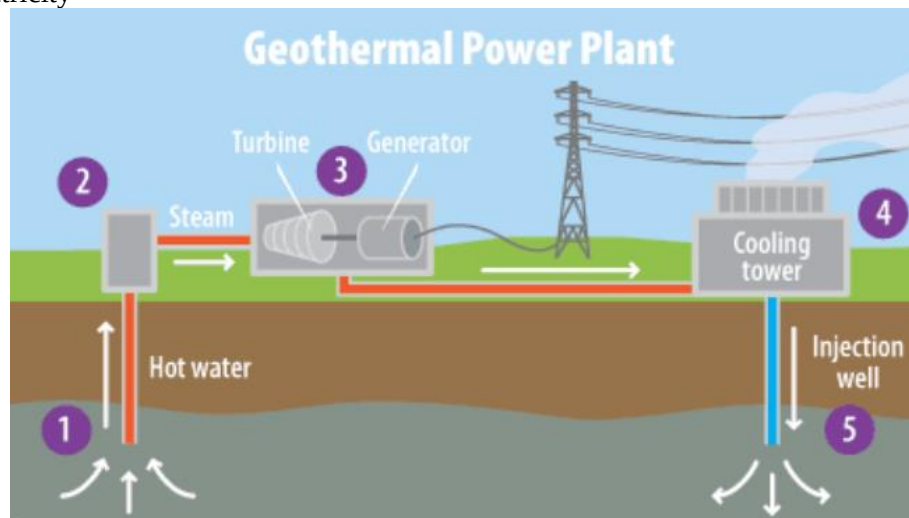
Geothermal heat pumps are used for space heating and cooling, as well as water heating. The earth (beneath the surface) remains at a relatively constant temperature throughout the year, warmer than the air above it during the winter and cooler in the summer. The geothermal heat pump takes advantage of this by transferring heat stored in the earth or in ground water into a building during the winter, and transferring it out of the building and back into the ground during the summer. The ground, in other

words, acts as a heat source in winter and a heat sink in summer. Generally, this kind of systems are employed along with the traditional systems to increase its efficiency



Electricity generation in a power plant

This is the most preferred method of geothermal energy harvesting, since electricity is the most versatile and useful form of energy. Geothermal power plants are used to convert the heat to the electricity



Global And Indian Scenario

Geothermal power plants operated in at least 24 countries in 2010, and geothermal energy was used directly for heat in at least 78 countries. These countries currently have geothermal power plants with a total capacity of 10.7 GW, but 88% of it is generated in just seven countries: The United States, the Philippines, Indonesia, Mexico, Italy, New Zealand, and Iceland. The most significant capacity increases since 2004 were seen in Iceland and Turkey. Both countries doubled their capacity. Iceland has the largest share of geothermal power contributing to electricity supply (25%), followed by the Philippines (18%).

The number of countries utilizing geothermal energy to generate electricity has more than doubled since 1975, increasing from 10 in 1975 to 24 in 2004. In 2003, total geothermal energy supply was 20 MToE (metric Tonne Oil Equivalent), accounting for 0.4% of total primary energy supply in IEA member countries. The share of geothermal in total renewable energy supply was 7.1%. Over the last 20 years, capital costs for geothermal power systems decreased by a significant 50%. Such large cost reductions are often the result of solving the “easier” problems associated with science and technology improvement in the early years of development. India has reasonably good potential for geothermal; the potential geothermal provinces can produce 10,600 MW of power (but experts are confident only to the extent of 100 MW). But yet geothermal power projects have not been exploited at all, owing to a variety of reasons, the chief being the availability of plentiful coal at cheap costs. However, with increasing environmental problems with coal-based projects, India will need to start depending on clean and eco-friendly energy sources in future; one of which could be geothermal.

Environmental Impact

Geothermal energy is one of the most environment friendly energy sources. Even though it has some concerns on environment.

- Fluids drawn from the deep earth carry a mixture of gases, notably carbon dioxide (CO₂), hydrogen sulfide (H₂S), methane (CH₄), ammonia (NH₃), and radon (Rn). If released, these pollutants contribute to global warming, acid rain, radiation, and noxious smells. Stations that experience high levels of acids and volatile chemicals are usually equipped with emission-control systems to reduce the exhaust. Geothermal Energy Geothermal stations could theoretically inject these gases back into the earth, as a form of carbon capture and storage.
- In addition to dissolved gases, hot water from geothermal sources may hold in solution trace amounts of toxic chemicals, such as mercury, arsenic, boron, antimony, and salt. These chemicals come out of solution as the water cools, and can cause environmental damage if released. The modern practice of injecting geothermal fluids back into the Earth to stimulate production has the side benefit of reducing this environmental risk.

Geothermal Energy Research

NREL's geothermal analysts, scientists, and engineers perform a full range of research from developing advanced methods of locating and accessing geothermal resources to exploring innovative technologies to extract and produce clean energy from the heat of the earth. Conducting analyses to determine the viability of geothermal energy production and likelihood of economic success. Exploring the potential benefits of combining geothermal with other renewable energy systems. Working to reduce barriers and accelerate deployment and implementation of geothermal technologies. Researching new approaches in geothermal assessment, decision making, and drilling. Designing new models and studying new techniques to increase the production of geothermal energy.

Future Possibilities

The reason that geothermal is expected to play an important role in the future is that we're getting better and better at doing this: we're now drilling geothermal wells with increasing efficiency, allowing more energy to be captured in each plant. The reason that geothermal is expected to play an important role in the future is that we're getting better and better at doing this: we're now drilling geothermal wells with increasing efficiency, allowing more energy to be captured in each plant. But the biggest looming technology is called Enhanced Geothermal Systems. See, as of now, only around 10% of the world's area is fit for geothermal power production. That's because you need hydrothermal convection systems -- places where hot water or steam bubbles up to the surface then sinks back down. If we're going to truly take advantage of the potential of geothermal, we'll need to begin injecting water deeper down, in 'hot dry rock' areas. Hydrothermal (or hot water) resources arise when hot water and/or steam is formed in fractured or porous rock at shallow-to-moderate depths (100 m to 4.5 km). **Enhanced Geothermal System (EGS)** is a man-made reservoir, created where there is hot rock but insufficient or little natural permeability or fluid saturation. Geopressured geothermal resources consist of hot brine, saturated with methane, found in large, deep aquifers under high pressure. The water and methane are trapped in sedimentary formations at a depth of about 3 km to 6 km, and the temperature of the water is in the range of 90C to 200C. Three forms of energy can be obtained from geopressured resources: thermal energy, hydraulic energy from the high pressure, and chemical energy from burning the dissolved methane gas.

TIDAL ENERGY RESOURCES

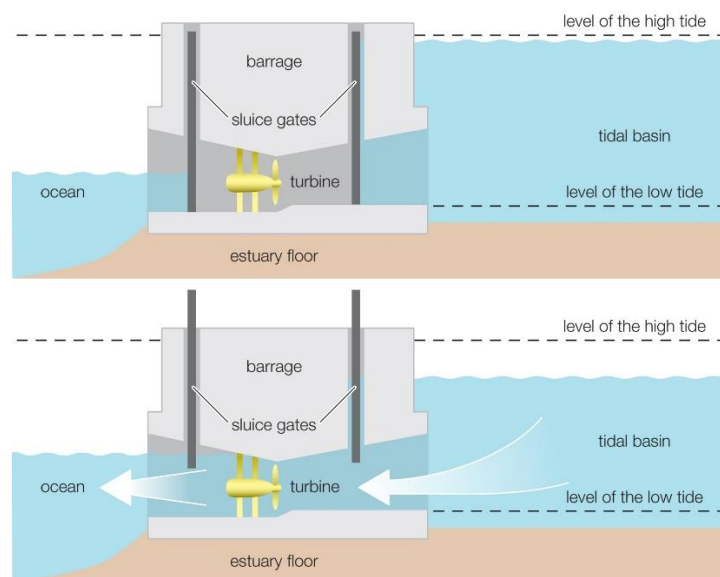
WHAT IS TIDAL ENERGY?

Tidal energy is form of hydropower that converts energy from the natural rise and fall of the tides into electricity. Tides are caused by the combined effects of the gravitational forces exerted by the sun, moon and rotation of the earth. Tidal plants can only be installed along the coastlines. Coastlines experience 2 high tides and 2 low tides on a daily basis. The difference in the water levels must atleast be 5meters to produce electricity.

DIFFERENT TIDAL TECHNOLOGIES

Tidal energy can be created from several technologies, main ones being:

- **Tidal turbines:** For most tidal energy generators, turbines are placed in tidal streams. As the tide moves through the turbine it imparts thrust on the blades causing a rotary motion. The turbine then generates electricity with the help of the turbines. Placing turbines in tidal streams is complex, because the machines are large and disrupt the tide they are trying to harness. The environmental impact could be severe, depending on the size of the turbine and the site of the tidal stream.
- **Tidal Barrage:** They are the most efficient tidal energy sources. A tidal barrage is a dam that utilizes the potential energy generated by the height between high and low tides Barrages can be constructed across tidal rivers, bays, and estuaries. At high tide, the wicket gates open. The water then moves through the turbines. The environmental impact of a barrage system can be quite significant. The land in the tidal range is completely disrupted. The change in water level in the tidal lagoon might harm plant and animal life. The salinity inside the tidal lagoon lowers, which changes the organisms that are able to live there.
- **Tidal lagoons:** The final type of tidal energy generator involves the construction of tidal lagoons. A tidal lagoon is a body of ocean water that is partly enclosed by a natural or manmade barrier. Tidal lagoons (have not been implemented yet due to their high construction cost) might also be estuaries and have freshwater emptying into them. A tidal energy generator using tidal lagoons would function much like a barrage. Unlike barrages, however, tidal lagoons can be constructed along the natural coastline. A tidal lagoon power plant could also generate continuous power. The turbines work as the lagoon is filling and emptying.



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ADVANTAGES OF TIDAL ENERGY

1. It is renewable and green

Tidal energy's source is a result of the effects of the sun and moon's gravitational fields, combined with our planet's rotation around its axis, which result in low and high tides. With this in mind, the power source of tidal energy is potentially renewable, whether we are talking about tidal barrages, stream generators or the more recent technology, dynamic tidal power (DTP). Compared to nuclear reserves and fossil fuels, the sun and moon's gravitational fields, as well as the Earth's rotation around its axis, will not cease to exist any time soon. Aside from being renewable, tidal energy is also an environmentally friendly energy source because it does not take up a lot of space and does not emit any greenhouse gases. However, there are already some examples of tidal power plants and their effects on the environment. Important studies and assessments are being conducted on these things.

2. It is predictable.

Sea currents are highly predictable, developing with well-known cycles, which makes it easier to construct tidal energy systems with the correct dimensions, since the kind of power the equipment will be exposed to is already known. This is why both the equipment's installed capacity and physical size have entirely other limitations, though tidal turbines and stream generators that are being used are very similar to wind turbines.

3. It is effective at low speeds.

Water is a thousand more dense than air, which makes it possible to produce electricity at low speeds. Based on calculations, power can be generated even at 1 minute per second, which is equivalent to a little over 3 feet per second.

4. It has a long lifespan.

So far, there is no reason to believe that tidal energy plants are not long lived. This means an ultimate reduction of the money spent on selling the electricity, making this energy source a very cost-competitive one. As an example, the La Rance tidal barrage power plant was constructed in 1966 and is still generating large amounts of electricity up to this day.

5. It reduces foreign importation of fuel.

By harnessing tidal energy on a large scale, we can help reduce foreign fuels importation and enhance energy security, as people would no longer have to rely much on foreign fuel import to satisfy the growing energy demand.

6. It serves as coastal protection.

Small dams and barrages, which are used to harness tidal energy, could protect ship ports and coastal areas from the dangerous tides during storms and bad weather conditions.

DISADVANTAGES OF TIDAL ENERGY

1. It still has some environmental effects.

As previously mentioned, tidal power plants are suspected to have some environmental effects, but are yet to be determined. As we know it, these facilities generate electricity with the use of tidal barrages that rely on ocean level manipulation, thus potentially having the same environmental effects as hydroelectric dams. Also, the turbine frames may potentially disrupt the natural movement of marine animals, and the construction of the whole plant may also disturb migration of fish. Nevertheless, technological solutions are now being developed to resolve these issues.

2. It is an intermittent energy source.

Tidal energy is considered as an intermittent source of energy, as it can only provide electricity when the tide surges, which happen about approximately 10 hours per day on average. This means that tidal energy can only be considered as reliable when accompanied with effective energy storage solutions.

3. It should be close to land.

Tidal energy facilities need to be constructed close to land, which is also the place where technological solutions that come with them are being worked on. It is hoped that in a few years we will be able to use weaker tidal currents at locations further out to sea. In addition to this disadvantage, the areas where this energy is produced are far away from the exact locations where it is consumed or needed.

4. It is expensive.

We should know that the method of generating tidal energy is relatively a new technology. It is projected that it will be commercially profitable by 2020 in larger scales with better technology. Also, the plants

that harness this type of energy are linked to higher upfront costs that are required for construction. Thus, tidal energy displays a lack of cost-effectiveness and efficiency in the world's energy market.

5. It is still considered a new technology.

Still a more theoretic source of power, tidal energy is limited in real life to just a few prototype projects because the technology has just begun to develop and needs plenty of research and huge funds before it reaches the commercial status.

6. It requires long gestation time.

The time and cost overruns can be huge for tidal power plants, which led to some of them being cancelled, such as UK's Severn Barrage. In fact, some tidal power stations, like the one being planned in Russia, will never be realized because of very long Gestation time.

WORLD SCENARIO

La Rance Tidal Power Station (240 MW) is the world's first large-scale tidal power plant in France that became operational in 1966. The power station is located on the estuary of the Rance River, in Brittany, France. It is also the second largest tidal power station in the world, In terms of installed capacity, after the Sihwa Lake Tidal Power Station in South Korea. Other countries also include UK, Canada, Russia, Netherlands and China.

INDIAN SCENARIO

Tidal Energy is still in Research & Development (R&D) stage. India has a long coastline with the estuaries and gulfs where tides are strong enough to move turbines for electrical power generation. The Gulf of Cambay and the Gulf of Kutch in Gujarat on the west coast have the maximum tidal range of 11m and 8m with average tidal range of 6.77m and 5.23m respectively. The Ganges Delta in the Sundarbans is approximately 5m with an average tidal range of 2.97m. No tidal power generation plant has been installed in India due to its high cost of generation of electricity and lack of techno economic viability of Tidal Energy Projects. Government of india estimates the identified economic power was about 7000MW in gulf of Cambay, about 1200MW in gulf of Kachchh in Gujarat and 100MW in gangetic delta.

ECONOMICS AND FEASIBILITY

Total cost of all marine technologies consists of investment and operation cost. Tidal Energy has an expensive initial cost (capital cost is very high) which may be one of the reasons tidal energy is not currently a popular source of renewable energy and therefore not economically viable. Although they are relatively new and actually the total cost is not established. However, the cost relation between each technology is visible.

The highest investment cost has barrage technology (620 million Francs). The most expansive part is barrage. It depends from the dam length and height. Planned the Severn Barrage power plant will has 214 with total capacity about 8,6GW (40MW each turbine). The generation covers 4% of Brittany houses. The total cost of 10-mile-long barrage with all required equipment is about 30 billion of US\$. The Rance power plant annual generation equals 600 million kWh (7200 hours) and planned the Sever power plant 17 billion kWh. However, the operational cost is relatively low, about €0,02 per kWh. It is related with inexpensive maintenance. One constructed power plant does not generate additional costs. Moreover, the barrage technology has long operational life time, even to 200 years.

The TSG technology is still in experimental stage and the final cost of the TSG power plant is not established. An investment cost depends from location. For installation at shoreline it is about 5650 €/kW, for near-shore 6825 €/kW and for offshore (in relatively long distance from shore) 8000 €/kW. Estimate components of the TSG project costs consist of:

- **mechanical and electrical components (about 30% of total cost),**
- **material and structure (about 20% of total cost),**
- **sitting, environmental monitoring and permitting (about 20% of total cost),**
- **mooring and foundation (about 15% of total cost),**

- installation and decommissioning (about 10% of total cost),
- transmission and grid connection (about 5% of total cost)

The last one point does not encompass cases when the grid is not strong enough. If the electrical network is weak there are additional cost related with grid development. Operation and maintenance costs are also depends from the location however it is not the only one factor which has influence on that costs

ENVIRONMENTAL IMPACTS

- Changing tidal flows by damming a bay or estuary could, however, result in negative impacts on aquatic and shoreline ecosystems, as well as navigation and recreation. Local tides changed only slightly due to the La Rance barrage, and the environmental impact has been negligible.
- However, the most publicized impact of tidal barrages is the potential loss of certain habitats. Installation of tidal plans are known to affect the aquatic life and changes in sedimentation processes in oceans.
- Tidal power can have effects on marine life. The turbines can accidentally kill swimming sea life with the rotating blades, although projects such as the one in Strangford feature a safety mechanism that turns off the turbine when marine animals approach. However, this feature causes a major loss in energy because of the amount of marine life that passes through the turbines.

To summarise many recent technological developments and improvements, both in design (e.g. dynamic tidal power, tidal lagoons) and turbine technology (e.g. new axial turbines, cross flow turbines), indicate that the total availability of tidal power may be much higher than previously assumed and that economic and environmental costs may be brought down to competitive levels

SOLAR ENERGY

Solar energy simply can be defined as energy provided by the sun's radiation where the sun is a very powerful source of energy. The sun is vast nuclear power plant of the fusion variety which generates power in the form of radiant energy at a rate of 3.8×10^{23} kW. An extremely small fraction of this is intercepted by earth, but even this small fraction amounts to the huge quantity of 1.8×10^{14} kW.

Solar Energy Benefits

1. Renewable Energy Source

Being a truly renewable energy source. It can be harnessed in all areas of the world and is available every day. We cannot run out of solar energy, unlike some of the other sources of energy. Solar energy will be accessible as long as we have the sun, therefore sunlight will be available to us for at least 5 billion years.

2. Reduces Electricity Bills

Since you will be meeting some of your energy needs with the electricity your solar system has generated, your energy bills will drop. How much you save on your bill will be dependent on the size of the solar system and your electricity or heat usage.

3. Diverse Applications

Solar energy can be used for diverse purposes like generating electricity (photovoltaics) or heat (solar thermal). It can be used to produce electricity in areas without access to the energy grid, to distill water in regions with limited clean water supplies and to power satellites in space.

4. Low Maintenance Costs

Solar energy systems generally don't require a lot of maintenance. You only need to keep them relatively clean, so cleaning them a couple of times per year is enough. Also, as there are no moving parts, there is no wear and tear.

5. Technology Development

Improvements in technology in the solar power industry is constantly advancing rapidly. Innovations in quantum physics and nanotechnology can potentially increase the effectiveness of solar panels and the electrical input of the solar power systems.

TECHNOLOGIES

The main types of technologies include solar thermal, solar heating, solar concentration and photovoltaic cells.

SOLAR THERMAL

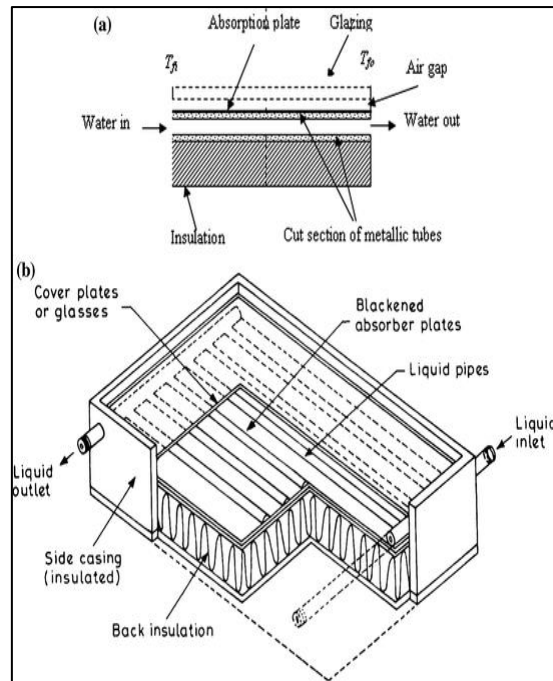
Solar thermal technologies involve harvesting energy from the sun for heating water or producing electrical power. Solar collectors are used for this purpose. They are mainly of two types:

1.Flat -plate collectors

A typical flat-plate collector is a metal box with a glass or plastic cover (called glazing) on top and a dark-coloured absorber plate on the bottom. The sides and bottom of the collector are usually insulated to minimize heat loss.

Working:

Sunlight passes through the glazing and strikes the absorber plate, which heats up, changing solar energy into heat energy. The heat is transferred to liquid passing through pipes attached to the absorber plate. And the liquid is used to govern the turbine for producing electricity.



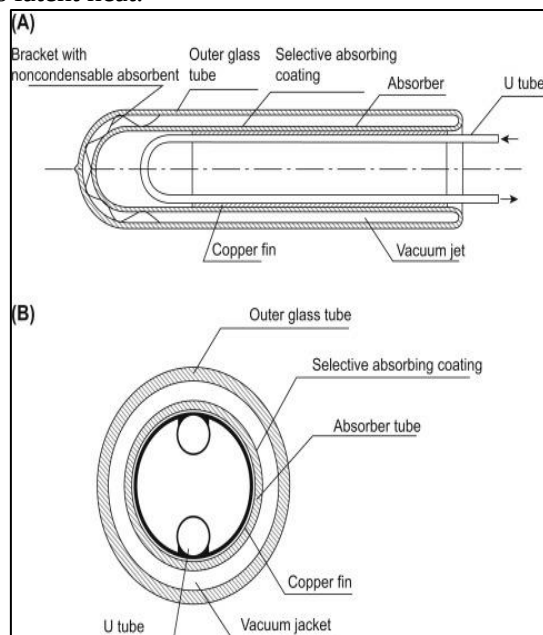
2. Evacuated-tube collectors

Evacuated tube collector (ETC) consists of single tubes that are connected to a header pipe. To reduce heat losses of the water-bearing pipes to the ambient air, each single tube is evacuated.

The ETCs provide the combined effects of a highly selective surface coating and vacuum insulation of the absorber element so that they can have high heat extraction efficiency compared with FPCs (flat plate collectors) in the temperature range above 80°C .

Working:

These collectors feature a heat pipe placed inside a vacuum-sealed tube. The pipe, which is a sealed copper pipe, attached to a black copper fin that fills the tube is a metal tip attached to the sealed pipe (condenser). The heat pipe contains a small amount of fluid that undergoes an evaporating–condensing cycle. In this cycle, solar heat evaporates the liquid, and the vapor travels to the heat sink region where it condenses and releases its latent heat.



SOLAR HEATING

Solar heating technologies collect the thermal energy from the sun and use this heat to provide hot water, space heating, cooling, and pool heating for residential, commercial, and industrial applications. These technologies displace the need to use electricity or natural gas.

1. Solar water heater

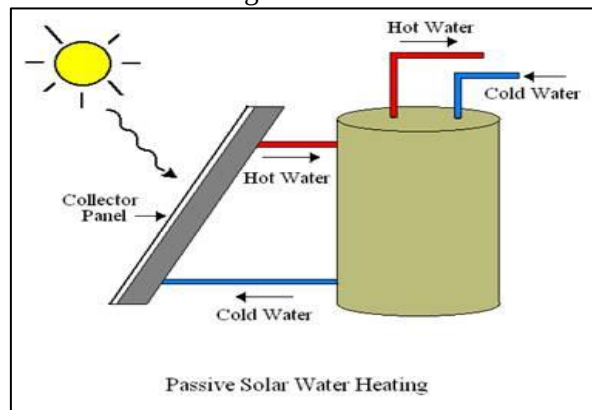
The conversion of sunlight into heat for water heating using solar thermal collector. There are two system for heating water: -

Passive system

Passive heating relies on architectural design to heat buildings. The building's site, structure, and materials can all be utilized to maximize the heating (and lighting) effect of the sunlight falling on it, thereby lowering or even eliminating its fuel requirement. A well-insulated building with a large glass window facing south, for instance, can effectively trap heat on sunny days and reduce reliance on gas or oil (for heating) or electricity (for lighting). Entering sunlight warms the air and the solid surfaces in those rooms exposed to it, and this warmth is carried to other rooms in the building

by natural convection. Interior finishes such as brick or tile are often incorporated into buildings to absorb the sunlight and reradiate the heat at night.

Passive systems are usually less expensive but less efficient too. They can be either integral collector/storage systems or thermosyphon systems. The integral collector/storage type is typically used to preheat water for a conventional water heater, and is best-suited to climates where temperatures seldom fall below freezing. Thermosyphon systems rely more on natural structure.



Active System

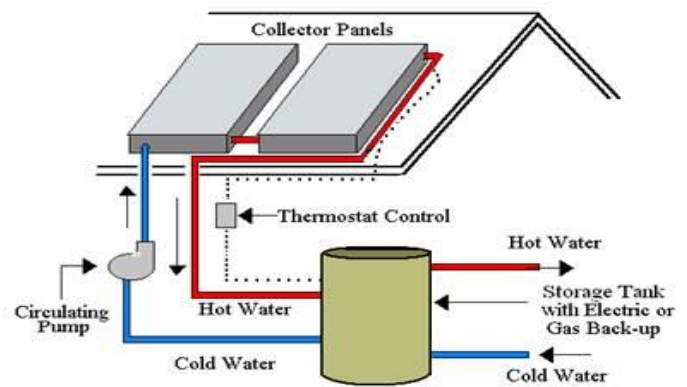
In active heating, mechanical means are used to store, collect, and distribute solar energy in buildings in order to provide hot water or space heating. The sunlight falling on a building's collector array is converted to heat, which is transferred to a carrier fluid (usually a liquid, less commonly air) that is then pumped to a conversion, storage, and distribution system. In liquid-based systems, water (or less commonly glycol) is pumped through tubes that are in contact with a flat-plate collector. The latter is a blackened metal plate that absorbs sunlight and is insulated on the front with layers of glass and air; the glass allows visible light to fall on the plate but traps the resulting heat, which is then transferred to the carrier fluid. Alternatively, the fluid may be pumped through

an evacuated glass tube or a volume of space onto which a large volume of sunlight has been focused (and hence concentrated) by reflecting mirrors.

After picking up heat from the collector, the carrier fluid is pumped down to an insulated storage tank, where it can be used immediately or stored for later use. The system can supply a home with hot water drawn from the storage tank, or, with the warmed water flowing through tubes in floors and ceilings, it can provide space heating. The storage tank allows water heated during sunny periods to be used at night or during cloudy days. If the carrier fluid contains antifreeze to keep it from freezing during cold weather, a heat exchanger is used to transfer the carrier fluid's heat to water that can

be used for domestic purposes. Residential heating systems using flat-plate collectors typically heat carrier fluids to temperatures between 66° C and 93° C (150° and 200° F). Active systems can be either direct circulation or indirect circulation.

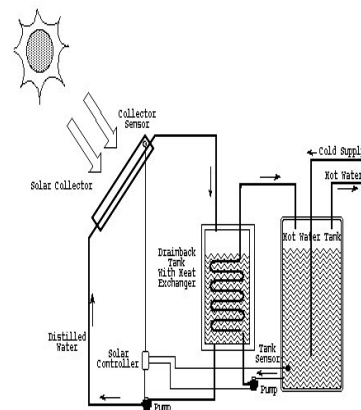
Direct circulation systems circulate domestic water through the collectors and to the storage tank. These are best-suited for mild climates where temperatures seldom drop below freezing. Indirect circulation systems circulate a non-freezing heat transfer fluid through the collectors and then through a heat exchanger in the storage tank.



Active Solar Water Heating System

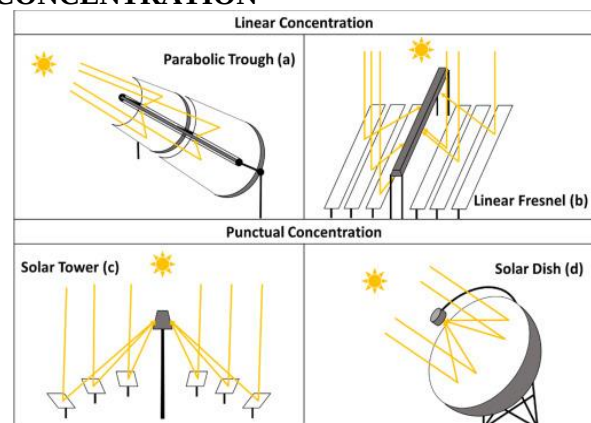
SPACE HEATING

A solar space heating system consist of an active system, a passive system or combination of both. Passive system is typically less costly & less than active systems. However, when retrofitting a building, active systems might be the only option for obtaining solar energy.



ELECTRICITY GENERATION USING SOLAR CONCENTRATION

Concentrating Solar Power (CSP) technologies use mirrors to concentrate (focus) the sun's light energy and convert it into heat to create steam to drive a turbine that generates electrical power. The plants consist of two parts: one that collects solar energy and converts it to heat, and another that converts the heat energy to electricity. CSP technology utilizes three alternative technological approaches: trough systems, power tower systems, and dish/engine systems.



PHOTOVOLTAIC CELLS

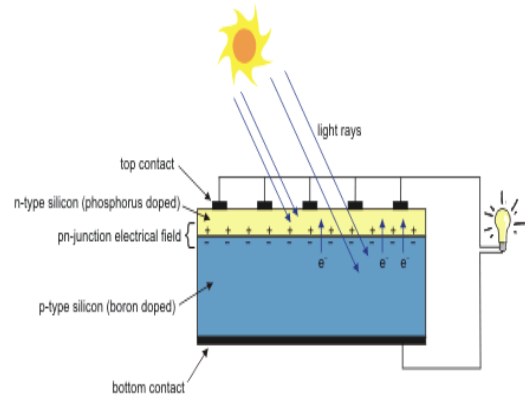
Photovoltaic cells are an integral part of solar-electric energy systems. It's a semiconductor diode which converts solar energy into electricity. Generally, silicon semiconductor is used.

Working

Photovoltaic cells need to establish an electric field. Much like a magnetic field, which occurs

due to opposite poles an electric field occurs when opposite charges are separated. To get this field, manufacturers "dope" silicon with other materials, giving each slice of the sandwich a positive or negative electrical charge, and this doped semiconductor use to produce electricity due to motion of electron &

hole present in it, & the motion starts because of solar energy.



Solar Power - Economics

The U.S. is the leading consumer of renewable energy. Yet, despite the increase of available solar energy over the past 10 years, solar still only accounts for 0.4% of total energy used in the U.S. Solar power also makes up 4% of consumption in the U.S.

Understanding the Cost of Solar Energy

In comparison to conventional hydrocarbon fuels such as coal or oil in generating electricity, the cost of solar energy is significantly higher. To compare energy cost, a framework is developed to measure energy costs by converting costs to kilowatt-hours.

Energy Costs

1 ton of coal costs \$36 = \$0.006 per KWH.

1 barrel of oil costs \$70 = \$0.05 per KWH

1 cubic foot of gas \$0.008 = \$0.03 per KWH

Solar Energy Costs

Average system costs = \$95 per square foot

Average solar panel output = 10.6 watts per square foot

Average solar energy system costs = \$8.95 per watt (to be converted into KWH)

When considering the economic viability of a solar panel system, it does not just involve the acquisition costs. The solar PV systems also create running costs during the period under examination. As a rough estimate, the operating costs can be assumed by between **1 to 1.5 percent** of the acquisition cost of the solar panel system, annually.

Solar power inverters

Solar power inverters are not as persistent as the solar PV modules. Thus, corresponding reserves must be saved. With a life expectancy of about **ten years** and costs between **10% and 20%** of the total price, approximately **1.5 %** of the total amount will fall due annually, which should be set aside for the replacement of the solar power inverter.

Maintenance costs of the solar panel system

The maintenance costs for the solar panel system constitute 1.5 percent. The costs are about **\$200 per year**.

Solar panel cleaning

If the removal of leaves, dust, flower pollen and other debris from solar PV modules is performed by a solar specialist company, then the costs are **\$5 to \$6** per solar module. The frequency of solar panel cleaning depends on regional pollution.

Insurance of the solar panel system

Depending on the size of the solar PV system, a special insurance makes sense. Liability insurance, as well as "All Risks" insurance of PV systems, insure against losses in the feed-in remuneration and also against high repair costs.

Indian Scenario:

The country's solar installed capacity reached 33.730 GW as of 31 December 2019. The Indian government had an initial target of 20 GW capacity for 2022, which was achieved four years ahead of schedule. In 2015 the target was raised to 100 GW of solar capacity (including 40 GW from rooftop solar) by 2022, targeting an investment of US\$100 billion. India has established nearly 42 solar parks to make land available to the promoters of solar plants.

India expanded its solar-generation capacity 8 times from 2,650 MW on 26 May 2014 to over 20 GW as on 31 January 2018. The country added 3 GW of solar capacity in 2015–2016, 5 GW in 2016–2017 and over 10 GW in 2017–2018, with the average current price of solar electricity dropping to 18% below the average price of its coal-fired counterpart. By the end of September 2019, India has installed more than 82,580 MW of renewable energy capacity with around 31,150 MW of capacity under various stages of installation. Power generation has reached 39.27 TWh by 2018-19.

Karnataka is the state leading in terms of installed solar power capacity (5.1GW) followed by Telangana, Rajasthan, Andhra Pradesh and Tamil Nadu.

Challenges and Opportunities:

- The land price is costly for acquisition in India.
- The architecture best suited to most of India would be a set of rooftop power-generation systems connected via a local grid.
- It is considered prudent to encourage solar-plant installations up to a threshold (such as 7,000 MW) by offering incentives.

GLOBAL SCENARIO

The disappearance of the fossil fuels from the planet and their rising costs has forced countries to switch over to a sustainable energy source. The large magnitude of solar energy available makes it a highly appealing source of electricity. Today, the technology produces less than one tenth of one percent of global energy demand. Countries like Japan, Germany, China and the United States are major markets for solar cells.

By the end of 2016, cumulative photovoltaic capacity increased by more than 75 gigawatt (GW) and reached at least 303 GW, sufficient to supply approximately 1.8 percent of the world's total electricity consumption. The top installers of 2016 were China, the United States, and India. There are more than 24 countries around the world with a cumulative PV capacity of more than one GW. Austria, Chile, and South Africa, all crossed the 1 GW-mark in 2016. Spain leads the world in terms of the concentrated solar power producing countries (2300 MW), USA (1738 MW), India (200 MW), South Africa (180 MW), UAE (100 MW), Algeria (25 MW) and other countries.

Researches Regarding Solar Energy

Goals:

- Improve efficiency
- Improve overall cost and cost-per-kWh
- Reduce impact of materials used
- Improve viability in less sunny conditions

The future of Solar Energy considers only the two widely recognized classes of technologies for converting solar energy into electricity, that is, photovoltaics (PV) and concentrated solar power (CSP), in their current and future forms. For the more than one billion people in the developing world who lack access to a reliable electric grid, the cost of small-scale PV generation is often outweighed by the very high value of access to electricity for lighting and charging mobile telephone and radio batteries.

Environmental Impacts

1. Land use

Depending on their location, larger utility-scale solar facilities can raise concerns about land degradation and habitat loss. Total land area requirements varies depending on the technology, the topography of the site, and the intensity of the solar resource.

2. Water use

In all manufacturing processes, some water is used to manufacture solar PV components. According to estimates that the Earth receives over 173,000 terawatts of energy every year, which is more than 10,000 times what humanity needs.

3. Hazardous materials

The PV cell manufacturing process includes a number of hazardous materials, like hydrochloric acid, sulfuric acid, nitric acid, hydrogen fluoride, 1,1,1-trichloroethane, and acetone.

4. Life-cycle global warming emissions

Emissions associated with other stages of the solar life-cycle, including manufacturing, materials transportation, installation, maintenance, and decommissioning and dismantlement.

Conclusion

Solar energy is clearly the future. The sun deploys more energy to the planet's surface than what's used every year. While the costs have reduced drastically over the years, the technology has remained the same. Researchers across the globe are working tirelessly to improve the way sunrays are collected and converted into energy. The relentless drive of technology will eventually help solar energy contribute a major part in the annual energy needs. Better and more efficient devices will be powered by the sun and have the ability to store this energy for longer periods.

WIND ENERGY

Wind energy (or **wind power**) describes the process by which wind is used to generate electricity. Wind turbines convert the kinetic energy in the wind into mechanical power. A generator can convert mechanical power into electricity

Wind is caused by the uneven heating of the atmosphere by the sun, variations in the earth's surface, and rotation of the earth. Mountains, bodies of water, and vegetation all influence wind flow patterns.

Benefits of using Wind Energy

Wind energy is plentiful, renewable, widely distributed, clean energy and it reduces greenhouse gas emissions if used to replace fossil-fuel-derived electricity. But a technology which offers remarkable advantages is not used to its full potential.

The major benefits include:

1. Wind energy produces no greenhouse gases.
2. Wind power plants can make a significant contribution to the regional electricity supply and to power supply diversification.
3. A very short lead time for planning and construction is required as compared to conventional power projects.
4. Wind energy projects are flexible with regard to an increasing energy demand - single turbines can easily be added to an existing park.
5. Finally, wind energy projects can make use of local resources in terms of labour, capital and materials.
6. Wind energy contributes to the power supply diversification,
7. Wind projects reinforce the cooperation with different donors including Germany, enhancing local capacities and technological know-how,
8. Wind projects attract new capital and can be included in the new approach of Independent Power Production

Technology

Wind turbines work on a simple principle: instead of using electricity to make wind—like a fan—wind turbines use wind to make electricity. Wind turns the propeller-like blades of a turbine around a rotor, which spins a generator, which creates electricity.

A wind turbine turns wind energy into electricity using the aerodynamic force from the rotor blades, which work like an airplane wing or helicopter rotor blade. When wind flows across the blade, the air pressure on one side of the blade decreases. The difference in air pressure across the two sides of the blade creates both lift and drag. The force of the lift is stronger than the drag and this causes the rotor to spin. The rotor connects to the generator, either directly (if it's a direct drive turbine) or through a shaft and a series of gears (a gearbox) that speed up the rotation and allow for a physically smaller generator. This translation of aerodynamic force to rotation of a generator creates electricity.

The majority of wind turbines consist of three blades mounted to a tower made from tubular steel. There are fewer common varieties with two blades, or with concrete or steel lattice towers. At 100 feet or more above the ground, the tower allows the turbine to take advantage of faster wind speeds found at higher altitudes.

Turbines catch the wind's energy with their propeller-like blades, which act much like an airplane wing. When the wind blows, a pocket of low-pressure air forms on one side of the blade. The low-pressure air pocket then pulls the blade toward it, causing the rotor to turn. This is called lift. The force of the lift is much stronger than the wind's force against the front side of the blade, which is called drag. The combination of lift and drag causes the rotor to spin like a propeller.

Another key component is the turbine's controller, that keeps the rotor speeds from exceeding 55 mph to avoid damage by high winds. An anemometer continuously measures wind speed and transmits the

data to the controller. A brake, also housed in the nacelle, stops the rotor mechanically, electrically or hydraulically in emergencies.

Types of wind turbines

There are two basic types of wind turbines: those with a horizontal axis, and those with a vertical axis. The majority of wind turbines have a horizontal axis: a propeller-style design with blades that rotate around a horizontal axis. Horizontal axis turbines are either upwind (the wind hits the blades before the tower) or downwind (the wind hits the tower before the blades). Upwind turbines also include a yaw drive and motor -- components that turn the nacelle to keep the rotor facing the wind when its direction changes.

While there are several manufacturers of vertical axis wind turbines, they have not penetrated the utility scale market (100 kW capacity and larger) to the same degree as horizontal axis turbines. Vertical axis turbines fall into two main designs:

- Drag-based, or Savonius, turbines generally have rotors with solid vanes that rotate about a vertical axis.
- Lift-based, or Darrieus, turbines have a tall, vertical airfoil style (some appear to have an eggbeater shape).

Parts of wind-energy turbine

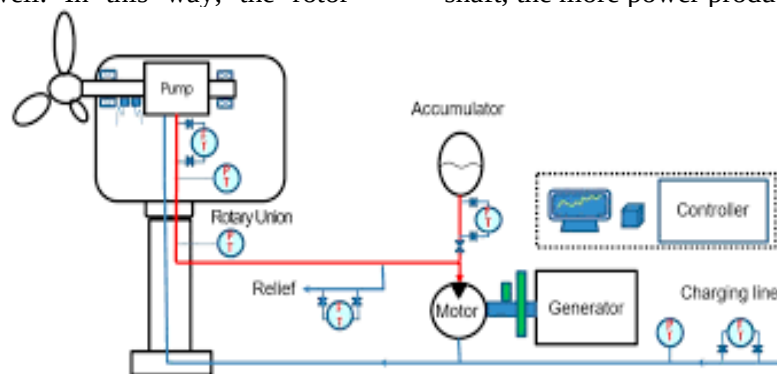
Rotor blades - The blades are basically the sails of the system; in their simplest form, they act as barriers to the wind (more modern blade designs go beyond the barrier method). When the wind forces the blades to move, it has transferred some of its energy to the rotor.

Shaft - The wind-turbine shaft is connected to the center of the rotor. When the rotor spins, the shaft spins as well. In this way, the rotor

transfers its mechanical, rotational energy to the shaft, which enters an electrical generator on the other end.

Generator - At its most basic, a generator is a pretty simple device. It uses the properties of electromagnetic induction to produce electrical voltage - a difference in electrical charge.

Tower - Tall tubular metal shaft. The taller the shaft, the more power produced



Cost calculation

- A typical 600kW turbine cost Rs 3.5 crore.
- Installation cost are typically Rs 92 lakhs.
- Therefore, total cost will be Rs 4.5 crore.
- The average price for large wind farm is around Rs 72k per kilowatt electrical power installed.
- Modern wind turbine is designed to work for some 1,20,000 hours of operation throughout their design lifetime of 20 years (13.7 year of nonstop).
- Maintenance cost are about 1-2 percent of total cost.

Present scenario

- Started development in 1990s.
- India has the 5th largest installed wind power capacity in the world.
- As of 31 jan 2017 the installed capacity of wind power in India was 19779.13 MW.
- Tamilnadu- 7158 MW (Generates 40 percent of India's wind power). Gujarat- 3093 MW (2nd in term of power generation). Maharashtra-2976 MW.

Demerits

- Firstly, constructing turbines and wind facilities is extremely expensive.
- The second disadvantage is technology immaturity.
- Wildlife: Wind turbines may be dangerous to flying animals. Many birds and bats have been killed by flying into the rotors. Experts are now conducting research to learn more about the effects that wind turbines have on marine habitats.
- Remoteness of location: Although this may be an advantage (placing wind turbines in desolate areas, far away from people), it may also be a disadvantage. The cost of travel and maintenance on the turbines increases and is time consuming. Offshore wind turbines require boats and can be dangerous to manage.
- Noise: Some wind turbines tend to generate a lot of noise which can be unpleasant